👩‍🏫 Staff Guide · Horticulture 1 · Unit 5 Soil & Nutrients (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in the class teacher's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: "Unit 5 Environmental Requirements for Good Plant Growth (textbook Unit 4): media and soil, N-P-K nutrients (definitions, excess, deficiency), deficiency photo research, soil texture jar test, self-evaluation questions")	**Level 2 piece**
Textbook Unit 4 + Plant Media and Nutrition	Slides "Plants Need", "Growing Media" + notes A (facts rewritten)
Soil Analysis Parts 1-2 (jar test)	Soil Jar Test lab packet + notes B
Plant Nutrition Definitions (N-P-K, excess, deficiency)	Slides + notes C
Deficiency photo research	Slide "Deficiency Signs"
Unit 4 SE questions (16)	Modified quiz
Class slides	Deck: 18 slides, Words to Know first, hold-up-1-2-3 checks, recap
Her worksheets (grouped into lessons)	Guided notes: 11 items + key
Class quiz / notes check	Quiz: 10 questions + key
Vocabulary	9 picture cards (cut apart)
—	1 game on the class site: 10 questions + a matching round

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no "all of the above"
- Jar test math: Level 1 says which layer is biggest; Level 2 uses a calculator frame
- Color anchors: 🟢 N, 🟣 P, 🟠 K

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the "look for" part

4. What to grade

- Soil analysis: Level 1 = photo + 'biggest layer'; Level 2 = layer % with calculator
- **Notes**: same points as her worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm in her files say "check with the class teacher".

🚧 Under construction (check with the class teacher)

- **Unit 4 SE question text**: The 16 self-evaluation questions are in the textbook scan only; our quiz is team-written.
- **Plant Media and Nutrition**: Title only in the export.

⚠️ Safety: tools and plants

- **Tools**: students may use all tools (pruners, knives, floral wire, low-temp glue guns) **with an adult right there**. Staff hand tools out, stay at the station, and count them back in. Cut-resistant gloves if possible.
- **Poinsettia sap** is milky and can irritate skin and eyes. Gloves on, no rubbing eyes, wash hands after.
- **Latex**: poinsettias are in the latex family. Students with a latex allergy wear non-latex (nitrile) gloves or skip handling; check health plans.
- **Bees and pollen**: check IEP/health plans for bee-sting or pollen allergies and know where the EpiPen is before greenhouse work.
- **Mouths**: no plants, soil, seeds, berries, fertilizer or rooting hormone in mouths. Students who mouth objects or have pica need one-to-one eyes in the greenhouse. Many plants (holly and mistletoe berries, some houseplants) are poisonous.
- **Chemicals**: rooting hormone and fertilizer with gloves, then wash hands. Pesticides and growth regulators are staff/teacher only; students never spray or handle them.
- **Greenhouse**: wet floors and hoses (slips), heat (water breaks), wet dry potting mix first (dust).

👩🏾‍🏫 Staff Guide · N-P-K Plant Check (Level 2)

Horticulture 1 · Unit 5 · goes with the class “Deficiencies / Plant Nutrition GH Participation” (4 pts). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Task card**: Modified from the class “Deficiencies / Plant Nutrition GH Participation”. 7 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 3 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- Show me the first step.
- Which tool do we need now?
- Where do our hands go when the tool is cutting?
- What should we do when we finish?

3. Modifications

- Task card instead of written directions
- Sentence starters on the reflection
- Partner reads directions aloud

4. What to grade (4 pts)

- Steps done (photo proof)
- Reflection: 2 sentences per box
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver (SLE).

🚧 Under construction (check with the class teacher)

- **This packet**: GH Participation nutrient activities were done in class and are not written in the export; this packet is team-written.

⚠️ Safety: tools and plants

- **Tools**: students may use all tools (pruners, knives, floral wire, low-temp glue guns) **with an adult right there**. Staff hand tools out, stay at the station, and count them back in. Cut-resistant gloves if possible.
- **Poinsettia sap** is milky and can irritate skin and eyes. Gloves on, no rubbing eyes, wash hands after.
- **Latex**: poinsettias are in the latex family. Students with a latex allergy wear non-latex (nitrile) gloves or skip handling; check health plans.
- **Bees and pollen**: check IEP/health plans for bee-sting or pollen allergies and know where the EpiPen is before greenhouse work.
- **Mouths**: no plants, soil, seeds, berries, fertilizer or rooting hormone in mouths. Students who mouth objects or have pica need one-to-one eyes in the greenhouse. Many plants (holly and mistletoe berries, some houseplants) are poisonous.
- **Chemicals**: rooting hormone and fertilizer with gloves, then wash hands. Pesticides and growth regulators are staff/teacher only; students never spray or handle them.
- **Greenhouse**: wet floors and hoses (slips), heat (water breaks), wet dry potting mix first (dust).

👩🏾‍🏫 Staff Guide · Soil Jar Test (Level 2)

Horticulture 1 · Unit 5 · goes with the class “Soil Analysis Parts 1-2” (8 pts). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Task card**: Modified from the class “Soil Analysis Parts 1-2”. 8 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 3 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- Show me the first step.
- Which tool do we need now?
- Where do our hands go when the tool is cutting?
- What should we do when we finish?

3. Modifications

- Task card instead of written directions
- Sentence starters on the reflection
- Partner reads directions aloud

4. What to grade (8 pts)

- Steps done (photo proof)
- Reflection: 2 sentences per box
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver (SLE).

⚠️ Safety: tools and plants

- **Tools**: students may use all tools (pruners, knives, floral wire, low-temp glue guns) **with an adult right there**. Staff hand tools out, stay at the station, and count them back in. Cut-resistant gloves if possible.
- **Poinsettia sap** is milky and can irritate skin and eyes. Gloves on, no rubbing eyes, wash hands after.
- **Latex**: poinsettias are in the latex family. Students with a latex allergy wear non-latex (nitrile) gloves or skip handling; check health plans.
- **Bees and pollen**: check IEP/health plans for bee-sting or pollen allergies and know where the EpiPen is before greenhouse work.
- **Mouths**: no plants, soil, seeds, berries, fertilizer or rooting hormone in mouths. Students who mouth objects or have pica need one-to-one eyes in the greenhouse. Many plants (holly and mistletoe berries, some houseplants) are poisonous.
- **Chemicals**: rooting hormone and fertilizer with gloves, then wash hands. Pesticides and growth regulators are staff/teacher only; students never spray or handle them.
- **Greenhouse**: wet floors and hoses (slips), heat (water breaks), wet dry potting mix first (dust).

Soil & Nutrients Notes

Horticulture 1 · Unit 5 Soil & Nutrients · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Media and soil

1. Greenhouses use potting _____ instead of garden soil.

2. Match:

Sand	smallest, sticky
Silt	balanced mix
Clay	largest, drains fast
Loam	medium, smooth

Word bank: **mix**

B. Jar test

3. The heaviest particles, _____, settle first.

4. Percent = layer ÷ _____ × 100.

Word bank: **sand** **total**

C. Nutrients

5. N = _____, green leaves.

6. P = _____, roots and flowers.

7. K = _____, strong stems.

8. Yellow older leaves can mean low _____.

Word bank: **nitrogen** **nitrogen** **phosphorus** **potassium**

Level 2 version of the gen ed “Plant Nutrition Definitions + Soil Analysis.”

📖 Soil & Nutrients: Vocabulary Picture Cards (Level 2)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Sand

The largest soil particles. Water drains fast through sand.



Silt

Medium-sized particles that feel smooth like flour.



Clay

The smallest particles. Clay is sticky and holds water.



Loam

A mix of sand, silt and clay that holds water and drains well.



Potting mix

A growing medium (peat, bark, perlite) used in greenhouse pots instead of garden soil.



Nitrogen (N)

Nutrient for green, leafy growth. Too little = yellow older leaves.



Phosphorus (P)

Nutrient for roots, flowers and seeds. Too little = purple leaves, poor roots.



Potassium (K)

Nutrient for strong stems and disease resistance. Too little = brown, scorched leaf edges.



Soil texture triangle

A triangle chart that uses % sand, silt and clay to name the soil type.

🟢 N-P-K Plant Check

Horticulture 1 · Unit 5 · Level 2



Name: _____

Modified from the class “Deficiencies / Plant Nutrition GH Participation”.

🧺 Supplies

- ☐ 🪴 **Greenhouse plants** staff choose
- ☐ 🃏 **N-P-K picture cards** yellow, purple, brown
- ☐ 🥄 **Fertilizer scoop** staff only measures
- ☐ 🧤 **Gloves**
- ☐ 📋 **Check chart**

🟢 Steps: check each box when you finish

- [ ] 1 🧤 Gloves on.
- [ ] 2 🪴 Inspect 4 plants, old and new leaves.
- [ ] 3 🔍 Match symptoms to N, P or K cards.
- [ ] 4 📋 Record plant, symptom and nutrient.
- [ ] 5 🔢 Read a fertilizer label: what are the N-P-K numbers?
- [ ] 6 🥄 Watch staff measure fertilizer; you water it in.
- [ ] 7 🧼 Wash hands.

🚧 **Under construction:** GH Participation nutrient activities were done in class and are not written in the export; this packet is team-written. *Check with the class teacher.*

🫙 Soil Jar Test

Horticulture 1 · Unit 5 · Level 2



Name: _____

Modified from the class “Soil Analysis Parts 1-2”.

🧺 Supplies

- ☐ 🫙 **Clear jar with lid**
- ☐ 🪱 **Soil** about 2 inches
- ☐ 💧 **Water**
- ☐ 📏 **Ruler**
- ☐ 🧤 **Gloves**
- ☐ 🧮 **Calculator** Level 2

🫙 Steps: check each box when you finish

1		Gloves on.	☐
2		Add 2 inches of soil; record the depth.	☐
3	💧	Fill with water to near the top.	☐
4	🔒	Lid tight; shake 10 seconds.	☐
5	⏳	Let settle overnight.	☐
6		Measure sand (bottom), silt (middle), clay (top).	☐
7		Percent = layer ÷ total × 100.	☐
8		Use the texture triangle to name the soil.	☐

🟢 N-P-K Plant Check · Prep Sheet

Horticulture 1 · Unit 5 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
🧑‍🌾 **Grower**	Does the main plant job with the tools	
🧺 **Supply helper**	Gets and returns supplies and tools to staff	
📸 **Recorder**	Takes photos and fills in the chart	
🧽 **Clean-up captain**	Sweeps, wipes tables, checks hands are washed	

🧰 Tools and supplies: check when you have it

- ☐ 🪴 **Greenhouse plants** staff choose
- ☐ 🃏 **N-P-K picture cards** yellow, purple, brown
- ☐ 🥄 **Fertilizer scoop** staff only measures
- ☐ 🧤 **Gloves**
- ☐ 📋 **Check chart**

⚠️ Safety and manners focus

🧤 Wear **gloves** for soil, sap, fertilizer and rooting hormone.

👄 **Nothing in your mouth**: no plants, soil, seeds or berries.

✂️ Sharp tools only with an **adult at the station**. Cut away from your body.

🥄 Staff measure fertilizer. **Never taste it.**

🧼 **Wash hands** when you finish.

✅ Before you start

☐ Gloves ☐ Apron ☐ Adult at station

🫙 Soil Jar Test · Prep Sheet

Horticulture 1 · Unit 5 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
🧑‍🌾 **Grower**	Does the main plant job with the tools	
🧺 **Supply helper**	Gets and returns supplies and tools to staff	
📸 **Recorder**	Takes photos and fills in the chart	
🧽 **Clean-up captain**	Sweeps, wipes tables, checks hands are washed	

🧰 Tools and supplies: check when you have it

- ☐ 🫙 **Clear jar with lid**
- ☐ 🪱 **Soil** about 2 inches
- ☐ 💧 **Water**
- ☐ 📏 **Ruler**
- ☐ 🧤 **Gloves**
- ☐ 🧮 **Calculator** Level 2

⚠️ Safety and manners focus

🧤 Wear **gloves** for soil, sap, fertilizer and rooting hormone.

👄 ****Nothing in your mouth****: no plants, soil, seeds or berries.

✂️ Sharp tools only with an ****adult at the station****. Cut away from your body.

😷 Dry soil dust: keep it low, don't breathe it in.

🧼 ****Wash hands**** when you finish.

✅ Before you start

☐ Gloves ☐ Apron ☐ Adult at station

🟢 N-P-K Plant Check · Reflection

Horticulture 1 · Unit 5 · Level 2



Name: _____

📸 My photos / proof

☐ Before photo ☐ During photo ☐ Finished photo

✅ Skills I used today (check all you did)

☐ 🧤 Used gloves and tools safely

☐ 🌱 Finished the N-P-K Plant Check steps

☐ 🧽 Cleaned up and returned tools

☐ 🤝 Did my job role

What did you do today, in order?

1. _____
2. _____

Ideas: **First…** **Next…** **Last…**

What did you learn about plants?

1. _____
2. _____

Ideas: **Plants need…** **I saw…** **Next time…**

🫙 Soil Jar Test · Reflection

Horticulture 1 · Unit 5 · Level 2



Name: _____

📸 My photos / proof

☐ Before photo ☐ During photo ☐ Finished photo

✅ Skills I used today (check all you did)

☐ 🧤 Used gloves and tools safely

☐ 🌱 Finished the Soil Jar Test steps

☐ 🧽 Cleaned up and returned tools

☐ 🤝 Did my job role

What did you do today, in order?

1. _____
2. _____

Ideas: **First…** **Next…** **Last…**

What did you learn about plants?

1. _____
2. _____

Ideas: **Plants need…** **I saw…** **Next time…**

✅ Soil & Nutrients Quiz

Horticulture 1 · Unit 5 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🪴 Greenhouses grow in potting mix because it…

A. Is heavy clay | B. Drains well and is clean | C. Has weed seeds

2. 🪨 The largest soil particle is…

A. Clay | B. Silt | C. Sand

3. 🧱 Clay soil…

A. Holds water and is sticky | B. Drains very fast | C. Is the biggest particle

4. 🌱 Loam is…

A. Pure sand | B. Fertilizer | C. A balanced mix of sand, silt and clay

5. 🫙 If sand is 1 inch of 2 inches total, sand is…

A. 50% | B. 10% | C. 100%

6. 🟢 Nitrogen helps…

A. Only flowers | B. Green leafy growth | C. Nothing

7. 🟣 Phosphorus helps…

A. Leaf color only | B. Roots, flowers and seeds | C. Pests

8. 🟠 Potassium helps…

A. Strong stems and disease resistance | B. Make soil sticky | C. Make leaves purple

9. 💜 Purple leaves and poor roots suggest low…

A. Nitrogen | B. Water | C. Phosphorus

10. 🧤 Handling fertilizer:

A. Gloves, never taste, wash hands | B. Bare hands are fine | C. Taste to test it

Soil & Nutrients Notes · ANSWER KEY

Horticulture 1 · Unit 5 Soil & Nutrients · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Media and soil

1. Greenhouses use potting **mix** instead of garden soil.

2. Match:

(Key: Sand → largest, drains fast · Silt → medium, smooth · Clay → smallest, sticky · Loam → balanced mix)

Sand	smallest, sticky
Silt	balanced mix
Clay	largest, drains fast
Loam	medium, smooth

Word bank: **mix**

B. Jar test

3. The heaviest particles, **sand**, settle first.

4. Percent = layer ÷ **total** × 100.

Word bank: **sand** **total**

C. Nutrients

5. N = **nitrogen**, green leaves.

6. P = **phosphorus**, roots and flowers.

7. K = **potassium**, strong stems.

8. Yellow older leaves can mean low **nitrogen**.

Word bank: **nitrogen** **nitrogen** **phosphorus** **potassium**

Level 2 version of the gen ed "Plant Nutrition Definitions + Soil Analysis."
Key written by our team from the class teacher's materials, please check.

✅ Soil & Nutrients Quiz · ANSWER KEY

Horticulture 1 · Unit 5 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🪴 Greenhouses grow in potting mix because it… Unit 4

- A. Is heavy clay
- **B. Drains well and is clean**
- C. Has weed seeds

2. 🪨 The largest soil particle is… Soil Analysis

- A. Clay
- B. Silt
- **C. Sand**

3. 🧱 Clay soil… Soil Analysis

- **A. Holds water and is sticky**
- B. Drains very fast
- C. Is the biggest particle

4. 🌱 Loam is… Soil Analysis

- A. Pure sand
- B. Fertilizer
- **C. A balanced mix of sand, silt and clay**

5. 🫙 If sand is 1 inch of 2 inches total, sand is… Soil Analysis

- **A. 50%**
- B. 10%
- C. 100%

6. 🟢 Nitrogen helps… Nutrition

- A. Only flowers
- **B. Green leafy growth**
- C. Nothing

7. 🟣 Phosphorus helps… Nutrition

- A. Leaf color only
- **B. Roots, flowers and seeds**
- C. Pests

8. 🟠 Potassium helps… Nutrition

- **A. Strong stems and disease resistance**
- B. Make soil sticky
- C. Make leaves purple

9. 💜 Purple leaves and poor roots suggest low… Deficiency

- A. Nitrogen
- B. Water
- **C. Phosphorus**

10. 🧤 Handling fertilizer: Safety

- **A. Gloves, never taste, wash hands**
- B. Bare hands are fine
- C. Taste to test it

Answers

1-B · 2-C · 3-A · 4-C · 5-A · 6-B · 7-B · 8-A · 9-C · 10-A

Reworded from the gen ed “Plant Nutrition + Soil Analysis”. Score: __ / 10. 🔑 Key written by our team from 's materials, please check.